

Residual Minimization And Scientific ML Reliability — cycles 16-17

2026-05-14

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Residual Minimization And Scientific ML Reliability — cycles 16-17

Abstract

Cycles 16-17 were procedural closure cycles. They did not add new scientific artifacts, proofs, simulations, validation runs, command outputs, or claims to the residual-minimization reliability package. Both cycles repeated the same conclusion: the substantive package had already converged, the final reader-facing artifact remained `reports/residual_minimization_reliability_final_report.md`, and there was no explicit trigger to reopen deferred scopes.

The auditor decision for the combined cycle range is **PIVOT**. In this context, PIVOT does not mean that the research package failed. It means the current cycle produced no new evidence and therefore should not receive a fresh validation label under the no-null-cycle rule. The validated package remains intact; the useful next step is handoff rather than further closure cycling.

Introduction

The research directive asked for a broad, grounded package on residual minimization in scientific machine learning. The target was not a generic critique of physics-informed neural networks, neural operators, collocation, or residual minimization. The target was a structured catalogue of limited-scope failure mechanisms, toy simulations, proof or obstruction notes, and certificates or corrected norms that explain when small residual loss does or does not certify physical correctness.

By the start of cycle 16, the package had already been reported as complete and validated. The relevant prior artifacts included:

- A broad residual-case catalogue with twenty-plus attempted mechanisms.
- Application-risk mapping across scientific machine-learning motifs.

- Branch notes for weak-topology failures, admissibility and invariant failures, ODE reliability, and collocation blind spots.
- A toy simulation suite with CSV, figure, and test coverage.
- A broad synthesis package.
- A final reader-facing report.

Cycles 16-17 did not reopen any of this work. Their purpose, as recorded in the researcher, worker, and auditor sessions, was to check whether any blocker or explicit reopening trigger existed. None was recorded.

Approach

This report consolidates the supplied cycle sessions and audit report. It does not re-audit the scientific claims or rerun validations.

The source records reviewed were:

Cycle	Role	Session ID	Date	Content
16	Researcher	4b638a5c-44b4-4f8e-a5e5-0420d4a72f39	2026-05-14T18:15:28Z	Procedural closure brief; stated that no new work, worker handoff, validation loop, or fanout was needed.
16	Worker	a76112e1-fefc-4728-a42b-e6fa8b7a71cf	2026-05-14T18:17:20Z	Reported that nothing was built and no commands were run.
16	Auditor	6c1a4785-e71b-474f-991a-4be3f5014bc2	2026-05-14T18:17:35Z	Returned PIVOT because the cycle added no artifact, proof, simulation, test result, or claim.
17	Researcher	e31dd355-e852-4bc5-8221-78a07a0e8ef3	2026-05-14T18:24:42Z	Repeated procedural closure assessment; no blocker or reopening trigger was present.

Cycle	Role	Session ID	Date	Content
17	Worker	7e71e4a1-c945-481c-8fd4-47d0a8cd0f0a	2026-05-14T18:24:58Z	Reported that nothing was built and no commands, simulations, edits, validations, or fanout were launched.
17	Auditor	2d0f86e8-071b-4c32-ab9e-c83a3a28daee	2026-05-14T18:25:13Z	Returned PIVOT for the same closure-only reason and identified handoff as the correct next step.

The supplied audit report for cycles 16-17 matches the session records. It lists no critical or moderate findings, only historical non-blocking warnings and documented deferred items.

Findings

Cycle 16 produced no new scientific work

Cycle 16 began with a researcher closure brief. The brief stated that the catalogue, application map, branch reports, toy simulation suite, synthesis package, and final report were already complete and validated. It also stated that M-2, M-4, CAT-18, and CAT-20 were pending or deferred items, not blockers.

The cycle 16 worker then reported that nothing was built. No commands were run, and no worker action, validation loop, or parallel fanout was launched.

The cycle 16 auditor returned PIVOT. The rationale was procedural: the worker output added no new scientific artifact, proof, simulation, test result, or claim. A closure-only cycle cannot receive fresh VALIDATED status under the no-null-cycle rule.

Cycle 17 repeated the same closure state

Cycle 17 repeated the same closure check. The researcher brief again stated that the package had converged and that no reopening trigger was present. The worker again reported that nothing was built and no commands were run.

The cycle 17 auditor also returned PIVOT. The auditor's sufficiency assessment stated that the directive's breadth criteria were already satisfied by prior validated milestones:

Criterion	Recorded status
Twenty-plus attempted mechanisms	Satisfied by validated M-7.

Criterion	Recorded status
Ten-plus explicit objective or function families	Satisfied by validated M-7 and M-12.
Multiple theorem-quality or rigorous obstruction branches	Satisfied by validated M-3, M-8, M-9, M-10, and M-12.
Five-plus toy simulations with data, figures, and tests	Satisfied by validated M-11.
Application-risk mapping	Satisfied by validated M-7 and M-12.
Final synthesis and reader-facing report	Satisfied by validated M-12 and _run/final-delivery-ready.
Explicit reopening trigger or new blocker	Absent.

PIVOT means handoff, not package failure

The PIVOT decision in these cycles is procedural. It means the cycle did not add evidence and should not be counted as a new validation event. It does not reverse earlier validations.

The final reader-facing artifact remains:

reports/residual_minimization_reliability_final_report.md

The relevant conclusion is that further closure-only cycles have no evidentiary value. A new research cycle would require either an explicit user request to reopen M-2, M-4, CAT-18, or CAT-20, or a concrete new blocker.

Historical warnings remain non-blocking

The cycles 16-17 audit report records only minor issues. These are historical non-blocking promise_check and organization warnings, including the _manager/validator-warnings entry at 2026-05-14T18:18:15Z. The audit report states that these warnings do not block the validated package.

The same audit report also preserves the documented loose ends:

- M-2: standalone continuous norm-mismatch theorem, pending or partly absorbed.
- M-4: standalone conservation-law selection campaign, pending or partly absorbed.
- CAT-18: inverse PDE or source sensor-nullspace case, deferred.
- CAT-20: long-horizon surrogate drift case, deferred.

These items remain limitations, not closure blockers.

Discussion

Cycles 16-17 add no new content to the scientific catalogue. Their value is procedural: they confirm that repeated closure checks should not be mistaken for new validation cycles. The package had already met the directive's breadth criteria through earlier validated artifacts, and neither cycle supplied a new reason to reopen research.

The practical consequence is narrow. The research package should be treated as complete for the current directive. Future work should be scoped explicitly to a deferred item or a new blocker rather than launched as another general closure cycle.

Open Questions

No new open questions were introduced in cycles 16-17.

The previously documented open or deferred items remain:

- M-2: standalone continuous norm-mismatch theorem.
- M-4: standalone conservation-law selection campaign.
- CAT-18: inverse PDE/source sensor-nullspace mechanism.
- CAT-20: long-horizon surrogate drift mechanism.

The source records state that these are not blockers for the validated residual-minimization reliability package.

References

- [1] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, 378, 686-707, 2019. <https://doi.org/10.1016/j.jcp.2018.10.045>
- [2] P. B. Bochev and M. D. Gunzburger, "Least-Squares Finite Element Methods," *Applied Mathematical Sciences* 166, Springer, 2009. <https://link.springer.com/book/10.1007/b13382>
- [3] A. Quarteroni and A. Valli, "Numerical Approximation of Partial Differential Equations," *Springer Series in Computational Mathematics* 23, Springer, 1994. <https://link.springer.com/book/10.1007/978-3-540-85268-1>
- [4] H. Wendland, "Scattered Data Approximation," *Cambridge Monographs on Applied and Computational Mathematics* 17, Cambridge University Press, 2005. <https://www.cambridge.org/core/books/scattered-data-approximation/966D4683385F92FB8D1535F8D0A9585E>
- [5] R. J. LeVeque, "Finite Volume Methods for Hyperbolic Problems," *Cambridge Texts in Applied Mathematics*, Cambridge University Press, 2002. <https://www.cambridge.org/core/books/finite-volume-methods-for-hyperbolic-problems/CB7B0A27A6D37AE3B906D4AE7C60A708>

Appendix: Implementation Details

Code organization

No code, proof note, simulation script, data file, or figure was created during cycles 16-17.

The manifest was refreshed before this report to include the previously rendered cycles 13-15 report:

- reports/cycles/report_cycles_13-15.md
- reports/cycles/report_cycles_13-15.pdf

This cycles 16-17 report is not included in the manifest counts below because the orchestrator writes it after this output block is parsed.

Test and command results

No scientific validation commands were run by the cycle 16 or cycle 17 worker sessions. Both workers explicitly reported that no commands were run.

The reporter used local session and file inventory commands only to consolidate the report and refresh MANIFEST.md. These reporter commands do not constitute new validation of the scientific package.

File counts from refreshed MANIFEST.md

Group	Files	Lines
Scripts/check scripts	12	1341
Tests	11	554
Residual-certificate notes	13	998
Final/report/cycle-report markdown files	8	1334
Root reference/ledger files	2	59
Total tracked files	46	4286

Session references

The session references used for this report are:

- Cycle 16 researcher: 4b638a5c-44b4-4f8e-a5e5-0420d4a72f39
- Cycle 16 worker: a76112e1-fefc-4728-a42b-e6fa8b7a71cf
- Cycle 16 auditor: 6c1a4785-e71b-474f-991a-4be3f5014bc2
- Cycle 17 researcher: e31dd355-e852-4bc5-8221-78a07a0e8ef3
- Cycle 17 worker: 7e71e4a1-c945-481c-8fd4-47d0a8cd0f0a
- Cycle 17 auditor: 2d0f86e8-071b-4c32-ab9e-c83a3a28daee

Cross-reference map

The relevant cross-reference chain remains unchanged from the validated package:

- residual-certificates/residual_case_catalogue.md records the broad mechanism catalogue.
- residual-certificates/application_risk_map.md links mechanisms to scientific machine-learning application motifs.
- residual-certificates/toy_simulation_results.md summarizes the representative toy simulations.
- residual-certificates/broad_synthesis_package.md integrates the validated catalogue, branches, toy evidence, certificates, stability baselines, and limitations.
- reports/residual_minimization_reliability_final_report.md is the final reader-facing report.
- promise_ledger.jsonl records M-12 validation at 2026-05-14T17:11:00Z and final-delivery validation at 2026-05-14T17:31:00Z.